

PneumaVision Knowledge Base

AI-Powered Chest X-Ray Analysis System

SECTION 1 — HOW TO USE PNEUMAVISION (Step-by-Step Guide)

What is PneumaVision?

PneumaVision is an AI-powered chest X-ray analysis platform. It uses a deep learning model (DenseNet121) trained on the NIH ChestX-ray14 dataset to detect 14 chest conditions from uploaded X-ray images. The system then generates a structured radiology-style report using Google Gemini AI, produces Grad-CAM heatmap visualizations showing which regions of the image the AI focused on, and allows patients to share reports with specialist doctors for review.

PneumaVision is a decision-support tool. It is not a replacement for clinical diagnosis. All AI findings should be reviewed by a qualified medical professional.

Feature 1 — Patient Registration and Login

How to Register

1. Visit the PneumaVision website and click **Sign In**.
2. Switch to the **Register** tab.
3. Enter your full name, gender, age, email address, and a password.
4. Click **Register**. Your account will be created immediately.

How to Log In

1. Enter your registered email and password on the **Sign In** tab.
2. Click **Login**. You will be redirected to the PneumaVision home page.

Doctor Login

1. Click the **Doctor** tab on the login page.
 2. Enter your assigned doctor credentials (email and password).
 3. You will be taken to the Doctor Portal with access to patient reports sent to you.
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Feature 2 — Uploading and Analysing a Chest X-Ray

Step 1 — Go to Analyse

After logging in, click **Analyse X-Ray** from the navigation bar or the home page button.

Step 2 — Upload Your Image

- Click the upload area or drag and drop your chest X-ray image.
- Accepted formats: **JPEG, JPG, PNG**
- Maximum file size: **20 MB**
- The image must be a real chest X-ray. The system will automatically reject non-medical images using AI validation.

Step 3 — Fill in Patient Details

Before analysing, fill in the patient information fields:

- **Patient Name** — your full name (auto-filled from your account)
- **Age** — your age in years
- **Sex** — Male or Female
- **Email** — your registered email (auto-filled)

Step 4 — Click Analyse X-Ray

The system will now run through 4 stages automatically:

1. **Uploading image** — the image is securely sent to the server
2. **Running DenseNet121 inference** — the AI model scores all 14 conditions
3. **Computing Grad-CAM heatmaps** — the AI highlights regions of interest
4. **Generating radiology report** — Gemini AI writes a structured report

This process typically takes 20–40 seconds.

Step 5 — View Your Results

Once analysis is complete, you will see:

- **Interactive Report tab** — findings with confidence levels, anatomical location, and priority
- **PDF Preview tab** — a printable radiology-style PDF

- **Images tab** — the original X-ray and the Grad-CAM heatmap overlay side by side
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Feature 3 — Understanding Your Report

Findings Section

Each detected condition is shown as a card with:

- **Condition name** — one of 14 possible chest findings
- **Confidence** — a percentage from 0–100% showing the AI's confidence
- **Anatomical Location** — which zone of the chest (e.g., Left Lower Zone)
- **Clinical Priority** — High (red), Moderate (orange), or Low (green)

Not Detected Section

Conditions the AI did not find above the detection threshold are listed here. This does not guarantee absence — low-quality or unusual X-rays may affect detection.

Impression

A short paragraph written by the AI summarising the overall findings, similar to a radiologist's impression.

Disclaimer

All reports include a disclaimer that PneumaVision is an AI tool and results must be confirmed by a qualified doctor.

Feature 4 — Grad-CAM Heatmap Visualisation

Grad-CAM (Gradient-weighted Class Activation Mapping) is a technique that highlights which parts of the X-ray image most influenced the AI's prediction for each detected condition.

- **Red/yellow areas** indicate regions the AI found most relevant
- **Blue/dark areas** indicate less relevant regions
- Each detected condition has its own heatmap panel
- Bounding boxes are drawn on the original image to show the approximate region of interest

This helps doctors quickly focus on the correct area when reviewing the AI findings.

Feature 5 — Downloading the PDF Report

On the results page, click **Download PDF** to save a professional radiology-style report. The PDF includes:

- Patient information table
- Original X-ray image
- Grad-CAM heatmap image
- Findings table with confidence and location
- Not-detected conditions
- Impression paragraph
- Disclaimer section

The PDF is suitable for sharing with healthcare providers.

Feature 6 — Sending Report to a Doctor

After analysis, click the **Send to Doctor** button. A dropdown will appear with the available specialist doctors. Select the appropriate doctor based on your condition or preference. The report will be sent to that doctor's portal for review.

You can also send past reports to doctors from the **My Reports** page.

Feature 7 — My Reports Page

Click **My Reports** in the navigation to view all your previous analyses. For each report you can:

- View the full interactive report and images
- Download the PDF again
- See doctor feedback if a doctor has reviewed your report
- Send the report to a different doctor

If a doctor has submitted feedback on your report, a green banner will appear showing their comments.

Feature 8 — Ask the PneumaVision Assistant (Chatbot)

A chatbot assistant is available on the analysis page. Click the **PneumaVision Assistant** button at the bottom of the screen to open it.

You can ask the assistant:

- Questions about any of the 14 chest conditions
- How to use PneumaVision features
- Information about the specialist doctors available
- General questions about what findings mean

The assistant uses Retrieval-Augmented Generation (RAG) — it searches a curated knowledge base to answer your questions accurately, rather than guessing.

Feature 9 — Report Chatbot (Ask About Your Report)

After analysis, a second chatbot is available specifically for your current report. Click **Ask about report** on the results page to open it.

You can ask questions like:

- "What does Atelectasis mean in my report?"
- "Which finding has the highest confidence?"
- "What is my clinical priority?"
- "Where exactly is the Effusion located?"

This chatbot has access to your specific report data and will only answer based on your results.

Feature 10 — Doctor Portal

Doctors log in through the Doctor tab on the login page. In the Doctor Portal, doctors can:

- See all patient reports sent to them, sorted by date
 - View the full report, both images, and AI findings
 - Submit written feedback to the patient
 - Correct AI labels (mark predictions as correct or incorrect and specify the true conditions)
 - Export labelled training data as a ZIP file for model retraining
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Frequently Asked Questions

Q: What file types are accepted? A: JPEG, JPG, and PNG files up to 20 MB.

Q: Can PneumaVision diagnose my condition? A: No. PneumaVision is an AI decision-support tool. All results must be reviewed and confirmed by a qualified doctor.

Q: What if the AI rejects my image? A: The system checks that the uploaded image is a real chest X-ray. If it rejects your image, make sure you are uploading a frontal chest radiograph (PA or AP view) in JPEG or PNG format.

Q: How long does analysis take? A: Typically 20–40 seconds depending on server load.

Q: Is my data stored securely? A: Your reports and images are stored in a secure database linked to your account email.

Q: Can I use PneumaVision without creating an account? A: No. Registration is required to save and access your reports.

Q: What happens after I send a report to a doctor? A: The doctor receives the report in their portal, reviews it, and may submit written feedback that you can see in My Reports.

SECTION 2 — SPECIALIST DOCTORS ON PNEUMAVISION

PneumaVision gives you access to four board-certified specialist doctors who review AI-generated chest X-ray reports and provide expert feedback.

Dr. Sarah Mitchell — Pulmonologist (Lung & Respiratory Specialist)

Email: sarah@pneumavision.com

Specialisation: Pulmonology — diseases of the lungs and respiratory system

About Dr. Mitchell: Dr. Sarah Mitchell is a senior pulmonologist with over 15 years of clinical experience in diagnosing and managing complex lung diseases. She completed her MD from Johns Hopkins University School of Medicine and completed her fellowship in Pulmonary and Critical Care Medicine at Massachusetts General Hospital, Boston. She has a special clinical interest in interstitial lung diseases, emphysema, pulmonary fibrosis, and chronic obstructive pulmonary disease (COPD).

Dr. Mitchell is particularly skilled at interpreting Grad-CAM heatmaps generated by AI systems and correlating AI findings with clinical presentation. She has reviewed over 8,000 chest X-rays throughout her career and regularly consults on complex pulmonology cases across multiple hospital networks.

Best suited for reports involving:

- Emphysema
- Pulmonary Fibrosis
- Atelectasis
- Pneumothorax
- Pleural Thickening
- Infiltration
- Consolidation
- Nodule

Languages: English

Response Time: Within 24 hours

Dr. James Anderson — Cardiologist (Heart & Cardiovascular Specialist)

Email: james@pneumavision.com

Specialisation: Cardiology — diseases of the heart and cardiovascular system

About Dr. Anderson: Dr. James Anderson is a consultant cardiologist with 13 years of experience in cardiac imaging, echocardiography, and heart failure management. He completed his MD from Harvard Medical School and his cardiology fellowship at Cleveland

Clinic, Ohio. He has a special interest in cardiomegaly, pulmonary oedema caused by cardiac dysfunction, and pleural effusions secondary to heart failure.

Dr. Anderson works closely with radiologists to interpret chest X-ray findings that have a cardiac origin. His training in advanced echocardiography and cardiac CT makes him well-equipped to contextualise AI-detected cardiovascular findings and provide clinically relevant guidance to patients and referring physicians.

Best suited for reports involving:

- Cardiomegaly
- Pulmonary Edema (Oedema)
- Pleural Effusion (cardiac cause)
- Consolidation (cardiac oedema)
- Infiltration (fluid overload)

Languages: English

Response Time: Within 24–48 hours

Dr. Priya Sharma — Radiologist (Chest Imaging Specialist)

Email: priya@pneumavision.com

Specialisation: Diagnostic Radiology — chest and thoracic imaging

About Dr. Sharma: Dr. Priya Sharma is a certified diagnostic radiologist specialising in thoracic and chest imaging. She completed her MBBS from Maulana Azad Medical College, Delhi, and her MD in Radiodiagnosis from KEM Hospital, Mumbai, followed by a fellowship in Thoracic Imaging at Stanford University Medical Center. With 11 years of experience, she has reported over 12,000 chest X-rays and CT scans across leading hospital settings.

Dr. Sharma has deep expertise in AI-assisted radiology and is the lead imaging reviewer on the PneumaVision platform. She is trained to identify subtle findings that AI models may over- or under-report, and focuses on providing clinically actionable imaging feedback. She is the recommended reviewer when findings are uncertain or multiple conditions are detected simultaneously.

Best suited for reports involving:

- All 14 conditions (general second-opinion imaging review)

- Uncertain or borderline AI findings
- Mass or Nodule evaluation
- Hernia (diaphragmatic / hiatal)
- Pleural Thickening
- Multiple simultaneous findings

Languages: English, Hindi

Response Time: Within 12–24 hours

Dr. Michael Chen — Infectious Disease Specialist

Email: michael@pneumavision.com

Specialisation: Infectious diseases — bacterial, viral, fungal, and tuberculosis-related lung infections

About Dr. Chen: Dr. Michael Chen is an infectious disease consultant with 14 years of clinical experience in pulmonary infections, tuberculosis, hospital-acquired pneumonia, and emerging respiratory pathogens. He completed his MD from Stanford University School of Medicine and his fellowship in Infectious Diseases at UCSF Medical Center, San Francisco. He is a recognised expert in managing complex pneumonia cases, TB-related lung changes, and post-viral lung complications.

Dr. Chen frequently reviews chest X-ray findings in patients with fever, productive cough, travel history, or immunocompromised status, helping correlate AI findings with infectious aetiology. He is the recommended reviewer for patients with suspected respiratory infections or conditions with an infectious cause.

Best suited for reports involving:

- Pneumonia
- Infiltration (infectious cause)
- Consolidation (infection-related)
- Nodule (tuberculosis-related)
- Pleural Effusion (parapneumonic or TB-related)
- Empyema-related findings

Languages: English, Mandarin

Response Time: Within 24 hours

How to Choose the Right Doctor

Your AI Finding	Recommended Doctor
Emphysema, Fibrosis, Atelectasis, Pneumothorax	Dr. Sarah Mitchell (Pulmonologist)
Cardiomegaly, Edema, Effusion (cardiac)	Dr. James Anderson (Cardiologist)
Nodule, Mass, Hernia, Multiple findings	Dr. Priya Sharma (Radiologist)
Pneumonia, Infiltration, Consolidation (infection)	Dr. Michael Chen (Infectious Disease)
Any report requiring general imaging review	Dr. Priya Sharma (Radiologist)

SECTION 3 — CHEST CONDITIONS DETECTED BY PNEUMAVISION

PneumaVision detects 14 chest conditions. Below is a clear, patient-friendly explanation of each.

1. Atelectasis

- What it is:** Atelectasis means partial or complete collapse of part of the lung. The tiny air sacs (alveoli) deflate or fail to expand, reducing oxygen exchange.
- How it looks on X-ray:** White or opaque areas where the lung should appear dark, reduced lung volume on one side, elevated diaphragm, crowded ribs.
- Common symptoms:** Shortness of breath, rapid breathing, persistent cough, chest tightness, low oxygen causing dizziness.
- Common causes:** Mucus plugging after surgery, tumours blocking airways, inhaled foreign objects, pressure from fluid or air outside the lung.
- When to see a doctor:** If you have new or worsening shortness of breath, persistent cough after surgery, or fever with breathing difficulty. Seek emergency care for sudden severe breathlessness or blue lips.

Recommended doctor on PneumaVision: Dr. Aisha Mehta (Pulmonologist)

2. Cardiomegaly

What it is: Cardiomegaly means an enlarged heart visible on chest X-ray. It is a sign of an underlying heart problem, not a disease by itself. The heart may be enlarged because it is working too hard, or the muscle has become weakened.

How it looks on X-ray: The heart shadow occupies more than normal chest width (cardiothoracic ratio over 50%), rounded or widened heart borders.

Common symptoms: Shortness of breath during activity, fatigue, leg or ankle swelling, palpitations, difficulty lying flat.

Common causes: High blood pressure, heart failure, cardiomyopathy, valve disease, severe anaemia, thyroid problems.

When to see a doctor: For ongoing breathlessness, new leg swelling, or frequent palpitations. Seek emergency care for chest pain, fainting, or sudden severe breathlessness.

Recommended doctor on PneumaVision: Dr. Rajiv Nair (Cardiologist)

3. Pleural Effusion

What it is: Pleural effusion is abnormal fluid buildup in the space between the lungs and chest wall. Even small amounts of extra fluid can compress the lung and reduce breathing capacity.

How it looks on X-ray: Blunting of the sharp angle at the base of the lung, white opacity at the lower lung zone, curved upper border of fluid rising along the chest wall.

Common symptoms: Shortness of breath, chest heaviness, dry cough, reduced exercise tolerance, sharp chest pain with breathing if the lining is inflamed.

Common causes: Heart failure (most common), pneumonia, tuberculosis, cancer, kidney disease, autoimmune disease, trauma.

When to see a doctor: For increasing breathlessness, cough lasting more than a few weeks, or fever with chest symptoms. Emergency care if severe breathlessness, blue lips, or sharp chest pain develops.

Recommended doctor on PneumaVision: Dr. Rajiv Nair (cardiac cause) or Dr. Sameer Joshi (infection-related)

4. Pulmonary Infiltrate

What it is: Infiltration means an area of the lung looks abnormally cloudy or dense on X-ray. This happens when fluid, pus, blood, inflammatory cells, or abnormal tissue fills part of the lung instead of air. It is a descriptive finding, not a specific diagnosis.

How it looks on X-ray: Patchy white or hazy areas in one or both lungs, ill-defined cloudy opacities, sometimes with visible air bronchograms.

Common symptoms: Cough (dry or with mucus), fever, shortness of breath, chest discomfort, fatigue.

Common causes: Bacterial pneumonia, viral infections, tuberculosis, aspiration, heart failure fluid, bleeding into lung tissue, drug reactions.

When to see a doctor: If cough persists more than a few weeks, fever with chest symptoms, or increasing breathlessness. Emergency care if severe breathlessness, coughing blood, or confusion.

Recommended doctor on PneumaVision: Dr. Sameer Joshi (infectious) or Dr. Rajiv Nair (fluid-related)

5. Pulmonary Mass

What it is: A pulmonary mass is an abnormal growth in the lung larger than 3 centimetres in diameter on imaging. Masses require urgent evaluation as they may be cancerous, infectious, or inflammatory in origin.

How it looks on X-ray: Rounded or irregular white density larger than 3 cm, may have smooth or irregular borders, sometimes with cavitation or associated lung collapse.

Common symptoms: Persistent cough, shortness of breath, chest discomfort, fatigue, unexplained weight loss, coughing blood.

Common causes: Primary lung cancer, cancer spread from other organs (breast, colon, kidney), tuberculosis granuloma, lung abscess, hamartoma (benign tumour).

When to see a doctor: Any abnormal chest X-ray showing a mass needs urgent evaluation. Emergency care if coughing significant blood, severe breathing difficulty, or collapse.

Recommended doctor on PneumaVision: Dr. Priya Sharma (Radiologist)

6. Pulmonary Nodule

What it is: A pulmonary nodule is a small spot in the lung, usually 3 centimetres or less. Most nodules are benign (healed infections or scar tissue), but some may represent early lung cancer. Follow-up imaging is often recommended.

How it looks on X-ray: Small round or oval white spot in the lung, may have smooth or irregular borders, may be single or multiple.

Common symptoms: Most nodules cause no symptoms. When symptoms occur: persistent cough, mild shortness of breath, recurrent infections in the same area.

Common causes: Healed tuberculosis or fungal infection, scar tissue, hamartoma, early lung cancer, cancer spread from another organ.

When to see a doctor: Any newly found nodule should be reviewed. Seek care for persistent cough, unexplained weight loss, or worsening symptoms. Emergency care if coughing blood or sudden severe breathlessness.

Recommended doctor on PneumaVision: Dr. Priya Sharma (Radiologist)

7. Pneumonia

What it is: Pneumonia is an infection of the lungs causing inflammation of the air sacs. The sacs fill with fluid, pus, and inflammatory material, reducing oxygen exchange. It ranges from mild illness to life-threatening disease.

How it looks on X-ray: Focal or patchy white areas in the lung, often in the lower lobes, lobar consolidation, air bronchograms (dark air-filled bronchi visible within white infected lung).

Common symptoms: Fever, chills, cough with mucus, shortness of breath, chest pain that worsens with breathing, fatigue, loss of appetite.

Common causes: Bacterial (*Streptococcus pneumoniae* most common), viral (influenza, coronaviruses), fungal, aspiration of food or vomit.

When to see a doctor: Fever with cough lasting more than a few days, shortness of breath, or worsening cough. Emergency care for severe breathing difficulty, blue lips, confusion, or inability to drink fluids.

Recommended doctor on PneumaVision: Dr. Sameer Joshi (Infectious Disease Specialist)

8. Pneumothorax

What it is: Pneumothorax is a collapsed lung caused by air leaking into the space between the lung and chest wall. This air reduces the pressure that keeps the lung expanded. It ranges from small and self-resolving to a life-threatening emergency (tension pneumothorax).

How it looks on X-ray: Visible thin line (pleural line) with no lung markings beyond it, increased blackness in the pleural space, possible shift of the heart and large vessels away from the affected side in severe cases.

Common symptoms: Sudden sharp one-sided chest pain, shortness of breath, chest tightness, rapid breathing, dry cough.

Common causes: Rupture of small air blisters on the lung surface (spontaneous), underlying COPD or emphysema, chest trauma, medical procedures.

When to see a doctor: Any sudden chest pain or unexplained shortness of breath needs evaluation. Emergency care for severe breathing difficulty, blue lips, rapid heart rate with dizziness, or fainting.

Recommended doctor on PneumaVision: Dr. Aisha Mehta (Pulmonologist)

9. Consolidation

What it is: Consolidation means lung tissue that has filled with fluid, pus, blood, or cells instead of air, making it denser and less able to exchange oxygen. The most common cause is pneumonia, but it can also be caused by oedema, aspiration, bleeding, or certain cancers.

How it looks on X-ray: Dense white area in a lung segment or lobe, visible air bronchograms, loss of normal boundaries between lung and nearby structures.

Common symptoms: Cough (dry or with mucus), fever, shortness of breath, rapid breathing, chest pain with breathing, fatigue.

Common causes: Bacterial or viral pneumonia, heart failure fluid, aspiration, bleeding into lung tissue, autoimmune lung inflammation.

When to see a doctor: Persistent cough or fever with breathing difficulty, worsening fatigue, or abnormal X-ray. Emergency care for severe breathing difficulty, coughing blood, blue lips, or confusion.

Recommended doctor on PneumaVision: Dr. Sameer Joshi (if infection) or Dr. Rajiv Nair (if cardiac)

10. Pulmonary Oedema (Edema)

What it is: Pulmonary oedema is excess fluid buildup inside the lungs, especially in the air sacs. Instead of air, the sacs fill with fluid, making oxygen exchange difficult. It is most commonly caused by heart failure.

How it looks on X-ray: Bilateral white haziness across both lungs, "bat-wing" pattern of cloudiness around the centre of the lungs, enlarged heart shadow, pleural effusions on both sides, fine linear markings (Kerley B lines).

Common symptoms: Shortness of breath (especially lying flat), waking gasping at night, leg swelling, persistent cough especially at night, fatigue. In severe acute oedema: pink frothy sputum, extreme breathlessness, blue lips.

Common causes: Heart failure (most common), kidney failure, severe infection, high altitude, toxic inhalation, drug overdose.

When to see a doctor: Increasing breathlessness, new difficulty lying flat, leg swelling, sudden weight gain, persistent cough at night. Emergency care for sudden severe breathlessness, pink frothy sputum, blue lips, or chest pain.

Recommended doctor on PneumaVision: Dr. Rajiv Nair (Cardiologist)

11. Emphysema

What it is: Emphysema is a chronic lung disease where the walls of air sacs break down over time, creating larger but less efficient air spaces. This traps air inside the lungs and makes it harder to breathe out. It is one of the main conditions included in COPD (Chronic Obstructive Pulmonary Disease) and is strongly linked to smoking.

How it looks on X-ray: Overexpanded lungs, flattened diaphragms, increased blackness from trapped air, widened spaces between ribs, reduced visibility of blood vessels in outer lung areas.

Common symptoms: Gradually worsening shortness of breath (especially with activity), chronic cough, wheezing, chest tightness, fatigue. Advanced: breathlessness at rest, barrel-shaped chest, low oxygen levels.

Common causes: Long-term smoking (main cause), secondhand smoke, occupational dust and fumes, air pollution, alpha-1 antitrypsin deficiency (genetic).

When to see a doctor: Increasing breathlessness, chronic cough, frequent chest infections, or reduced ability to exercise. Emergency care for sudden severe breathing difficulty, blue lips, sudden chest pain, or collapse.

Recommended doctor on PneumaVision: Dr. Aisha Mehta (Pulmonologist)

12. Pulmonary Fibrosis

What it is: Pulmonary fibrosis is a chronic condition where lung tissue becomes scarred, thickened, and stiff. Scarring makes it harder for lungs to expand and for oxygen to pass into the bloodstream. Unlike infections that heal, fibrosis usually represents permanent structural damage that may gradually worsen.

How it looks on X-ray: Reticular (net-like) or streaky white opacities, volume loss with smaller-appearing lungs, irregular scarring patterns predominantly in the lower and outer lung areas.

Common symptoms: Progressive shortness of breath (especially with exertion), dry persistent cough, fatigue, reduced stamina. Advanced: breathlessness at rest, low oxygen, need for supplemental oxygen, clubbed fingertips.

Common causes: Idiopathic (no known cause), autoimmune diseases (rheumatoid arthritis, scleroderma, lupus), asbestos or silica dust exposure, prior severe pneumonia or tuberculosis, radiation therapy, certain medications.

When to see a doctor: Increasing shortness of breath over weeks or months, dry persistent cough, reduced exercise tolerance, or abnormal X-ray. Emergency care for sudden worsening breathlessness, low oxygen, blue lips, or chest pain.

Recommended doctor on PneumaVision: Dr. Aisha Mehta (Pulmonologist)

13. Pleural Thickening

What it is: Pleural thickening is scarring or thickening of the pleura, the thin membrane lining the lungs and inner chest wall. When the pleura becomes thickened and stiff, lung movement may be restricted. It is usually a result of prior infection, injury, surgery, or long-term exposure to asbestos.

How it looks on X-ray: White linear or plaque-like thickening along the chest wall, blunted or irregular lung base angles, calcified plaques in asbestos-related disease, volume loss if extensive.

Common symptoms: Many people with mild pleural thickening have no symptoms. When present: mild shortness of breath, chest tightness, dull chest discomfort, reduced exercise tolerance, fatigue with activity.

Common causes: Prior tuberculosis, severe pneumonia involving the pleura, empyema, pleural effusion that has resolved, asbestos exposure, autoimmune disease, chest trauma or surgery.

When to see a doctor: New or worsening shortness of breath, persistent chest discomfort, ongoing dry cough, or abnormal X-ray finding. Emergency care for severe breathing difficulty, sharp chest pain, coughing blood, or rapid decline.

Recommended doctor on PneumaVision: Dr. Priya Sharma (Radiologist) or Dr. Aisha Mehta (Pulmonologist)

14. Hernia (Hiatal / Diaphragmatic)

What it is: In chest X-ray datasets, "hernia" refers to a hiatal hernia or diaphragmatic hernia — where part of the stomach or abdominal organs moves upward through an opening or weakness in the diaphragm into the chest cavity. Small hernias may cause no symptoms. Larger ones can cause reflux, swallowing difficulty, or breathing problems.

How it looks on X-ray: Air-fluid level behind the heart (from stomach in hiatal hernia), gas bubble above the diaphragm, irregular or elevated diaphragm contour, bowel loops visible in the chest in larger hernias.

Common symptoms: Heartburn or acid reflux, sour taste, regurgitation after meals, chest discomfort or burning, difficulty swallowing, belching, bloating, chronic cough, hoarseness. Large hernias: shortness of breath.

Common causes: Weakening of diaphragm tissue with age, obesity, repeated straining, pregnancy, chest trauma (diaphragmatic hernia), congenital defect (in infants).

When to see a doctor: Frequent heartburn, trouble swallowing, chest discomfort after meals, chronic cough, or abnormal X-ray. Emergency care for severe chest or abdominal pain, persistent vomiting, inability to swallow, or coughing blood.

Recommended doctor on PneumaVision: Dr. Priya Sharma (Radiologist)

This knowledge base is maintained by the PneumaVision team and is intended for patient education and system guidance. It is not a substitute for professional medical advice.

This knowledge base covers 14 chest conditions detectable on chest X-rays using the PneumaVision AI system: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia. Each section covers what the condition is, how it appears on X-ray, symptoms, causes, prevention, and when to seek medical care.

1. Atelectasis

What is it?

Atelectasis is the **partial or complete collapse of part of the lung**, causing reduced or absent air in the affected lung tissue. Normally, the lungs contain millions of tiny air sacs called alveoli that expand with each breath and allow oxygen to pass into the bloodstream. In atelectasis, these air sacs deflate or fail to fully expand, leading to decreased oxygen exchange. The condition may involve a small segment of lung, an entire lobe, or rarely a whole lung.

Atelectasis is common after surgery, especially chest or abdominal surgery, because pain and shallow breathing prevent full lung expansion. It can also occur when an airway becomes blocked by mucus, a tumor, or a foreign object, or when pressure from outside the lung (such as fluid or air in the pleural space) compresses the lung. Small areas of atelectasis may cause few symptoms, while larger collapse can become serious and require urgent treatment.

How is it detected on a Chest X-ray?

Chest X-ray is one of the most common ways to detect atelectasis. Radiologists look for signs of **volume loss** and increased density in the collapsed portion of lung.

Common findings include:

- White or opaque area where air-filled lung should normally appear darker
- Loss of normal lung volume in one segment or lobe
- Shift of the trachea, mediastinum, or hilum **toward** the affected side
- Elevated diaphragm on the involved side
- Crowding of ribs due to reduced lung expansion
- Displacement of fissures between lung lobes
- Plate-like linear opacities at the lung bases (subsegmental atelectasis)
- Compensatory overinflation of nearby healthy lung tissue

The pattern depends on which lobe is affected. CT scan may be used when the cause is unclear or to rule out tumors or mucus plugging.

Symptoms

Symptoms depend on how much of the lung is collapsed, how quickly it developed, and whether another disease is present. Small areas may cause no symptoms and may only be found on imaging. Larger or sudden atelectasis often causes noticeable breathing problems.

Common symptoms include:

- Shortness of breath or feeling unable to take a deep breath
- Rapid or shallow breathing
- Persistent cough
- Chest tightness or mild chest discomfort
- Wheezing in some cases
- Fatigue or reduced stamina
- Low oxygen levels causing dizziness or headache

More serious symptoms may include:

- Severe breathlessness
- Fast heart rate
- Bluish lips or fingertips (cyanosis)
- Anxiety or restlessness due to low oxygen
- Fever if infection develops behind blocked airways

After surgery, patients may mainly complain of pain when breathing deeply, inability to cough effectively, and worsening oxygen saturation.

Older adults may present with confusion, weakness, or sudden decline rather than obvious respiratory distress.

Causes & Risk Factors

Atelectasis occurs when air cannot reach part of the lung or when outside pressure compresses lung tissue.

Obstructive Causes (airway blockage)

These block airflow into the lung:

- Mucus plugs, especially after surgery or in bedridden patients
- Tumors inside the airway
- Inhaled foreign body (common in children)
- Blood clots or thick secretions
- Severe asthma causing blockage of smaller airways

Non-Obstructive Causes (external pressure or poor expansion)

These prevent lung expansion:

- Pleural effusion (fluid around lung)

- Pneumothorax (air around lung)
- Enlarged heart pressing nearby lung tissue
- Scarring from prior infection or lung disease
- Weak breathing effort due to pain, sedation, or neuromuscular disease

Important Risk Factors

- Recent surgery, especially chest or abdominal surgery
- General anesthesia
- Long bed rest or immobility
- Smoking
- Chronic lung disease such as COPD or asthma
- Advanced age
- Obesity
- Poor cough reflex
- Heavy sedation or opioid pain medications
- Neurological disease affecting breathing muscles
- Dehydration causing thick mucus

Hospitalized and ICU patients are at particularly high risk.

Prevention

Many cases of atelectasis—especially postoperative cases—can be prevented with good lung care.

After Surgery

- Use an incentive spirometer regularly
- Take deep breaths and cough as instructed
- Walk early and often after surgery
- Sit upright rather than lying flat continuously
- Ensure pain is controlled enough to breathe deeply

General Prevention

- Stop smoking, especially before surgery
- Stay hydrated to keep mucus thinner
- Treat asthma or COPD properly
- Avoid prolonged immobility

- Practice breathing exercises if bedridden
- Follow chest physiotherapy recommendations when prescribed

For Children

- Keep small objects away to prevent choking and airway blockage

For High-Risk Patients

People with chronic illness, weak cough, or prior lung disease may benefit from respiratory therapy and close monitoring during hospital stays.

When to see a doctor

Medical evaluation is recommended if breathing becomes difficult or symptoms persist, especially after surgery or illness.

See a doctor promptly if you have:

- New or worsening shortness of breath
- Persistent cough not improving
- Fever with cough after surgery
- Chest discomfort with breathing
- Wheezing or inability to clear mucus
- Ongoing fatigue with low oxygen readings

Seek urgent or emergency care if you develop:

- Sudden severe breathlessness
- Blue lips or fingertips
- Rapid breathing or inability to speak full sentences
- Severe chest pain
- Confusion or fainting
- Very low oxygen saturation
- Symptoms after choking or suspected inhaled object

People recovering from surgery, elderly individuals, and those with heart or lung disease should seek care early because atelectasis can worsen quickly or lead to pneumonia.

2. Cardiomegaly

What is it?

Cardiomegaly means an **enlarged heart** seen on imaging such as a chest X-ray. It is not a disease by itself, but rather a sign that the heart has increased in size due to another underlying condition. The enlargement may happen because the heart muscle becomes thickened, the chambers become dilated and stretched, or both. In some cases, the heart enlarges temporarily, while in others it reflects chronic heart disease.

The heart may enlarge when it has to work harder than normal, such as from long-standing high blood pressure, valve disease, or weakened heart muscle. Over time, the enlarged heart may pump less effectively, leading to heart failure, fluid buildup, abnormal heart rhythms, or reduced oxygen delivery to the body.

Some people with mild cardiomegaly have no symptoms and only discover it during routine imaging. Others may have significant symptoms depending on the cause and how well the heart is functioning.

How is it detected on a Chest X-ray?

Chest X-ray is one of the common first tests that may suggest cardiomegaly. Radiologists assess the size and shape of the heart shadow (cardiac silhouette) compared with the width of the chest.

Common findings include:

- Enlarged cardiac silhouette occupying more than expected chest width
- Increased **cardiothoracic ratio** (typically >50% on a PA chest X-ray in adults)
- Rounded or widened heart borders
- Globular heart shape in some conditions (such as pericardial effusion)
- Signs of heart failure such as lung congestion or pulmonary edema
- Pleural effusions in advanced cases

Portable AP chest X-rays may exaggerate heart size, so formal PA films are usually more accurate. Echocardiography is often used afterward to evaluate heart function, chamber size, valve disease, and pumping strength.

Symptoms

Symptoms vary depending on the underlying cause and whether the enlarged heart is functioning normally. Some people have no symptoms for years, while others develop progressive heart failure or arrhythmias.

Common symptoms include:

- Shortness of breath during activity
- Fatigue or low energy
- Reduced exercise tolerance
- Feeling weak or easily exhausted

- Swelling of ankles, feet, or legs
- Rapid heartbeat or palpitations
- Mild chest discomfort

Symptoms that may worsen over time:

- Shortness of breath while lying flat
- Waking at night gasping for air
- Persistent cough, especially when lying down
- Weight gain from fluid retention
- Abdominal bloating from fluid buildup
- Dizziness or near fainting

Serious symptoms include:

- Chest pain
- Severe breathlessness at rest
- Irregular heartbeat
- Fainting episodes
- Sudden swelling
- Confusion or poor circulation

Older adults may mainly report fatigue, weakness, reduced walking ability, or swelling rather than classic chest symptoms.

Causes & Risk Factors

Cardiomegaly develops when the heart must work harder or when heart muscle becomes diseased.

Common Causes

- **High blood pressure (hypertension):** forces the heart to pump harder, causing thickening and enlargement
- **Coronary artery disease:** reduced blood supply weakens heart muscle
- **Heart failure:** stretched and weakened chambers enlarge over time
- **Cardiomyopathy:** disease of the heart muscle, genetic or acquired
- **Heart valve disease:** leaking or narrowed valves increase workload
- **Arrhythmias:** long-standing rhythm disorders can weaken the heart
- **Congenital heart disease:** structural abnormalities present from birth
- **Pulmonary hypertension:** high pressure in lung circulation enlarges right heart

Other Medical Causes

- Severe anemia
- Thyroid disorders
- Kidney disease with fluid overload
- Chronic alcohol misuse
- Viral myocarditis
- Pregnancy-related cardiomyopathy
- Iron overload disorders

Risk Factors

- Smoking
- Diabetes
- Obesity
- High cholesterol
- Sedentary lifestyle
- Family history of heart disease
- High salt intake
- Untreated sleep apnea
- Older age

Prevention

Preventing cardiomegaly largely means preventing or controlling the conditions that strain the heart.

Heart-Healthy Lifestyle

- Maintain normal blood pressure
- Exercise regularly as medically appropriate
- Maintain healthy body weight
- Eat a balanced diet low in excess salt, trans fats, and processed foods
- Avoid smoking and tobacco exposure
- Limit alcohol intake
- Manage stress and sleep quality

Medical Prevention

- Control diabetes and cholesterol

- Take blood pressure medicines as prescribed
- Treat valve disease early when needed
- Manage thyroid or kidney disease
- Screen family history of inherited heart conditions
- Seek evaluation for persistent palpitations or breathlessness

High-Risk Individuals

People with hypertension, diabetes, obesity, or family history should have regular health checkups and cardiac screening when advised.

When to see a doctor

Medical evaluation is important if symptoms suggest heart strain or heart failure.

See a doctor promptly if you have:

- Ongoing shortness of breath with activity
- New leg or ankle swelling
- Persistent fatigue without clear reason
- Frequent palpitations
- Difficulty lying flat due to breathing discomfort
- Rapid weight gain from fluid retention
- Chronic cough worse at night

Seek urgent or emergency care if you develop:

- Chest pain or pressure
- Severe shortness of breath at rest
- Fainting or collapse
- Fast or irregular heartbeat with dizziness
- Sudden severe swelling
- Blue lips or signs of low oxygen
- Confusion or inability to stay awake

People with known heart disease should seek care early if symptoms worsen suddenly, as enlarged heart conditions can progress to serious heart failure or dangerous rhythm problems.

3. Effusion (Pleural Effusion)

What is it?

Pleural effusion is the **abnormal buildup of fluid in the pleural space**, the thin space between the lungs and the inner chest wall. Normally, only a very small amount of fluid is present there to help the lungs move smoothly during breathing. When too much fluid accumulates, it can compress the lung, reduce lung expansion, and make breathing difficult.

Pleural effusion is not a disease by itself but a result of another medical problem. It may develop gradually over days to weeks or appear rapidly depending on the cause. Small effusions may cause no symptoms, while large ones can significantly reduce oxygen levels and require urgent drainage.

The fluid may be watery, inflammatory, infected, bloody, or cancer-related depending on the underlying condition.

How is it detected on a Chest X-ray?

Chest X-ray is a common first test used to identify pleural effusion. Because fluid appears whiter than air-filled lung, radiologists look for characteristic fluid patterns.

Common findings include:

- **Blunting of the costophrenic angle** (loss of the sharp corner where diaphragm meets chest wall)
- White opacity at the lower lung zone or lung base
- Meniscus sign: curved upper border of fluid rising along the chest wall
- Partial hiding of the diaphragm or heart border
- Large effusions causing compression of the lung
- Shift of the mediastinum or trachea **away** from the affected side if very large
- Layering fluid on side-lying (decubitus) films

Small effusions may be subtle on X-ray. Ultrasound is often more sensitive and is commonly used to confirm fluid and guide drainage. CT scan may help determine the cause.

Symptoms

Symptoms depend on how much fluid is present, how quickly it developed, and the underlying disease causing it. Small pleural effusions may cause no symptoms and be found incidentally.

Common symptoms include:

- Shortness of breath, especially with exertion
- Feeling unable to take a full deep breath
- Chest heaviness or pressure
- Dry cough
- Reduced exercise tolerance

- Fatigue
- Discomfort when lying flat

If the pleura becomes inflamed, symptoms may include:

- Sharp chest pain that worsens with deep breathing or coughing
- Pain on one side of the chest
- Tenderness with movement

If infection is present:

- Fever
- Chills
- Sweating
- Productive cough
- General weakness

Large effusions may cause:

- Rapid breathing
- Low oxygen levels
- Severe breathlessness
- Anxiety or distress due to air hunger

Older adults may mainly complain of weakness, confusion, poor appetite, or decline in mobility rather than chest pain.

Causes & Risk Factors

Pleural effusion occurs when too much fluid is produced or not enough is removed from the pleural space.

Common Causes

Transudative Effusions (fluid imbalance)

Usually caused by pressure changes rather than inflammation:

- **Heart failure** (most common cause)
- Kidney disease
- Liver cirrhosis
- Low blood protein states
- Fluid overload

Exudative Effusions (inflammation/injury)

Usually caused by disease of the pleura or lung:

- Pneumonia (parapneumonic effusion)
- Tuberculosis
- Cancer involving lung or pleura
- Pulmonary embolism
- Autoimmune diseases such as lupus or rheumatoid arthritis
- Pancreatitis
- Trauma or surgery

Other Types

- **Hemothorax:** blood in pleural space
- **Chylothorax:** lymphatic fluid leakage
- **Empyema:** infected pus in pleural space

Risk Factors

- Known heart failure
- Smoking
- Chronic kidney or liver disease
- Lung cancer risk factors
- Recent pneumonia
- Tuberculosis exposure
- Autoimmune disease
- Recent chest surgery or trauma
- Older age

Prevention

Prevention focuses on managing the conditions that commonly lead to pleural effusion.

General Prevention

- Control heart failure with medicines and follow-up
- Manage blood pressure and heart disease
- Treat kidney and liver disease properly
- Avoid smoking
- Maintain healthy weight and exercise regularly

Infection Prevention

- Seek treatment early for pneumonia or persistent cough
- Vaccinations such as influenza and pneumococcal vaccines when appropriate
- Good hand hygiene
- Tuberculosis screening in high-risk settings

Cancer Risk Reduction

- Stop smoking
- Follow cancer screenings when recommended
- Seek care for unexplained weight loss or chronic cough

Hospital / Recovery Prevention

- Early mobilization after surgery
- Follow breathing exercises after chest or abdominal procedures
- Report worsening breathing symptoms quickly

When to see a doctor

Medical evaluation is important for unexplained breathing symptoms or chest discomfort.

See a doctor promptly if you have:

- Increasing shortness of breath
- Persistent cough lasting more than a few weeks
- Chest discomfort when breathing deeply
- Fever with cough
- Swelling of legs with breathing difficulty (possible heart failure)
- Unexplained fatigue or reduced stamina
- New symptoms in someone with cancer, kidney disease, or heart disease

Seek urgent or emergency care if you develop:

- Severe shortness of breath at rest
- Rapid breathing or inability to speak full sentences
- Blue lips or signs of low oxygen
- Sharp chest pain with worsening breathing difficulty
- High fever with shaking chills
- Confusion or fainting

- Sudden symptoms after chest injury

People with known heart failure, cancer, tuberculosis risk, or recent pneumonia should seek care early because pleural effusions can enlarge and sometimes require drainage or urgent treatment.

4. Infiltration (Pulmonary Infiltrate)

What is it?

“Infiltration” or “pulmonary infiltrate” is a **broad radiology term** used when an area of the lung appears abnormally dense or cloudy on a chest X-ray. It usually means that something other than normal air has filled or affected part of the lung tissue. This may include fluid, pus, blood, inflammatory cells, protein material, or abnormal tissue growth.

Importantly, infiltration is **not a specific diagnosis**. It is a descriptive imaging finding that suggests an underlying problem that needs further evaluation. Depending on the cause, infiltrates may be temporary and reversible, or they may reflect chronic lung disease.

An infiltrate may affect one small region, one lobe, both lungs, or appear as diffuse patchy changes throughout the lungs. Doctors interpret it together with symptoms, blood tests, medical history, and sometimes CT imaging.

How is it detected on a Chest X-ray?

Chest X-ray shows infiltrates as areas that look whiter or denser than normal air-filled lung. Radiologists assess the pattern, location, and extent to narrow possible causes.

Common X-ray findings include:

- Patchy white or hazy opacities
- Ill-defined cloudy areas in one or both lungs
- Localized infiltrate in one lobe (often infection)
- Bilateral diffuse infiltrates (possible edema, viral illness, inflammation)
- Interstitial reticular or streaky infiltrates
- Ground-glass type haziness (better seen on CT)
- Air bronchograms in some cases if alveoli are filled
- Associated pleural effusion or enlarged heart depending on cause

Patterns help guide diagnosis:

- **Lower lobe infiltrate:** common in pneumonia or aspiration
- **Both lungs:** may suggest edema or viral disease
- **Upper lobe infiltrate:** can suggest tuberculosis or chronic disease
- **Peripheral infiltrates:** sometimes seen in inflammatory or viral conditions

CT scan is often used when chest X-ray findings are unclear or symptoms are severe.

Symptoms

Symptoms depend entirely on the underlying cause. Some people have minimal symptoms, while others may be seriously ill.

Common symptoms include:

- Cough, dry or productive
- Fever or chills
- Shortness of breath
- Chest discomfort or pain
- Fatigue
- Weakness
- Wheezing in some cases
- Reduced exercise tolerance

If infection is present:

- Thick mucus or sputum
- Sweating
- Body aches
- Loss of appetite

If fluid overload or heart failure is present:

- Breathlessness when lying flat
- Swelling of legs
- Sudden nighttime shortness of breath

If bleeding or severe inflammation is present:

- Coughing blood
- Rapid breathing
- Low oxygen levels
- Severe fatigue

Older adults may present with confusion, falls, poor appetite, or sudden weakness instead of classic respiratory symptoms.

Causes & Risk Factors

Pulmonary infiltrates can result from many medical conditions.

Infectious Causes

- Bacterial pneumonia
- Viral respiratory infections
- Tuberculosis
- Fungal lung infections
- Aspiration pneumonia after choking or vomiting

Non-Infectious Causes

- Pulmonary edema from heart failure
- Lung hemorrhage (bleeding into lung tissue)
- Autoimmune inflammation
- Interstitial lung disease
- Drug reactions affecting lungs
- Radiation injury
- Acute respiratory distress syndrome (ARDS)

Other Causes

- Cancer spreading into lung tissue
- Eosinophilic lung disease
- Protein accumulation disorders

Risk Factors

Risk depends on the cause, but common factors include:

- Smoking
- Chronic lung disease (COPD, asthma)
- Weak immune system
- Diabetes
- Older age
- Recent hospitalization
- Bedridden state
- Difficulty swallowing / aspiration risk
- Heart disease
- Occupational dust or chemical exposure

Prevention

Because infiltration is a finding rather than a single disease, prevention means reducing risk of common underlying lung conditions.

Respiratory Infection Prevention

- Vaccinations (influenza, pneumococcal, others when recommended)
- Handwashing and infection control
- Avoid close exposure to contagious illness when possible
- Seek treatment early for persistent cough or fever

Lung Protection

- Avoid smoking and secondhand smoke
- Use protective masks around dust or chemicals
- Manage asthma or COPD properly
- Maintain physical activity and lung fitness

Aspiration Prevention

- Eat slowly if swallowing difficulty exists
- Sit upright during meals
- Follow swallowing therapy if recommended
- Avoid heavy sedation when possible

Heart and General Health

- Control blood pressure and heart failure
- Maintain healthy weight
- Manage diabetes
- Attend regular medical follow-ups

When to see a doctor

Because infiltrates may represent infection, heart failure, inflammation, or other serious illness, medical review is important when symptoms are present.

See a doctor promptly if you have:

- Persistent cough lasting more than a few weeks
- Fever with cough or chest discomfort
- Increasing shortness of breath
- Fatigue not improving

- New abnormal chest X-ray finding
- Unexplained weight loss
- Recurrent chest infections

Seek urgent or emergency care if you develop:

- Severe shortness of breath
- Rapid breathing or low oxygen readings
- Blue lips or fingertips
- Chest pain
- Confusion or fainting
- Coughing blood
- High fever with weakness
- Sudden worsening in elderly or medically fragile patients

People with heart disease, cancer, chronic lung disease, or weakened immunity should seek evaluation early because infiltrates may progress quickly or indicate serious underlying disease.

5. Mass

What is it?

A pulmonary mass is an **abnormal growth or lesion in the lung that is typically larger than 3 centimeters in diameter** on imaging. The term “mass” describes size and appearance rather than a specific diagnosis. A lung mass may be benign (non-cancerous), malignant (cancerous), infectious, inflammatory, or related to scar tissue.

Because lung masses are larger than nodules, they often receive more urgent evaluation, especially in adults with smoking history or other cancer risk factors. Some masses are discovered when a patient develops symptoms such as cough or weight loss, while others are found incidentally during routine imaging.

A lung mass can arise from lung tissue itself, the airways, nearby pleura, or spread from another part of the body. Further testing such as CT scan, PET scan, bronchoscopy, or biopsy is often needed to determine the cause.

How is it detected on a Chest X-ray?

Chest X-ray may show a pulmonary mass as a focal abnormal opacity or rounded density in the lung. Radiologists evaluate its size, shape, borders, and location.

Common X-ray findings include:

- Rounded or irregular white opacity larger than 3 cm

- Solitary or multiple masses
- Smooth borders (sometimes benign) or spiculated/irregular margins (more concerning)
- Upper lobe location in some smoking-related cancers
- Cavitation (central breakdown or air-filled area) in certain infections or cancers
- Associated collapse (atelectasis) if airway obstruction occurs
- Enlarged lymph nodes near the mediastinum or hilum
- Pleural effusion in advanced disease

Chest X-ray can suggest a mass, but **CT scan is usually required** for better characterization.

Symptoms

Some lung masses cause no symptoms early and are found incidentally. Symptoms often depend on size, location, and whether the lesion is cancerous or causing airway blockage.

Common symptoms include:

- Persistent cough
- Cough that changes from baseline in smokers
- Shortness of breath
- Wheezing localized to one area
- Chest discomfort or dull chest pain
- Fatigue
- Reduced exercise tolerance

Symptoms that may suggest more serious disease:

- Coughing blood (hemoptysis)
- Unexplained weight loss
- Loss of appetite
- Night sweats
- Recurrent pneumonia in the same area
- Hoarseness
- Swelling of face or neck (rare, due to vessel compression)

If advanced cancer is present:

- Bone pain
- Headaches or neurological symptoms

- Severe weakness
- Progressive breathlessness

Older adults may present with subtle symptoms such as fatigue, decreased appetite, or recurrent infections.

Causes & Risk Factors

A lung mass can have many causes, both benign and malignant.

Benign Causes

- Granulomas from prior infections (such as tuberculosis or fungal disease)
- Hamartoma (common benign lung tumor)
- Lung abscess
- Rounded pneumonia
- Inflammatory pseudotumor
- Scar tissue

Malignant Causes

- Primary lung cancer
- Metastatic cancer spread from breast, colon, kidney, melanoma, and others
- Lymphoma involving the chest

Infectious / Inflammatory Causes

- Tuberculosis
- Fungal infections
- Chronic inflammatory disease

Risk Factors for Cancerous Masses

- Smoking (strongest major risk factor)
- Secondhand smoke exposure
- Older age
- Family history of lung cancer
- Radon exposure
- Asbestos exposure
- Air pollution
- Prior radiation therapy to chest

- COPD or emphysema

Prevention

Not all lung masses can be prevented, but cancer risk can be significantly reduced.

Reduce Cancer Risk

- Do not smoke or stop smoking as early as possible
- Avoid secondhand smoke
- Test homes for radon where recommended
- Use occupational protection around asbestos, silica, or chemicals
- Maintain healthy diet and regular exercise

Prevent Infectious Causes

- Early treatment of tuberculosis or fungal disease when diagnosed
- Vaccination and infection prevention practices
- Prompt treatment of persistent lung infections

Screening

People at high risk for lung cancer (such as long-term smokers in eligible age groups) may benefit from **low-dose CT screening** as recommended by healthcare guidelines.

General Health

- Manage COPD or chronic lung disease
- Attend follow-up imaging when advised

When to see a doctor

Medical evaluation is important for any suspicious lung mass or ongoing respiratory symptoms.

See a doctor promptly if you have:

- Persistent cough lasting more than a few weeks
- Recurrent chest infections
- New shortness of breath
- Chest pain not improving
- Unexplained fatigue
- Unintentional weight loss
- Abnormal chest X-ray showing a mass

Seek urgent or emergency care if you develop:

- Coughing significant blood
- Severe breathing difficulty
- Rapid worsening chest pain
- Facial swelling or difficulty swallowing
- Confusion or collapse
- Severe weakness or dehydration

People with smoking history, prior cancer, unexplained weight loss, or coughing blood should seek evaluation quickly, as early diagnosis greatly improves treatment options.

6. Nodule

What is it?

A pulmonary nodule is a **small, rounded or irregular spot in the lung**, usually measuring **3 centimeters (30 mm) or less** in diameter on imaging. Nodules are common findings on chest imaging and are often discovered incidentally when a chest X-ray or CT scan is performed for another reason. Most pulmonary nodules are benign, but some may represent early lung cancer or another important disease.

A nodule can be caused by healed infection, inflammation, scar tissue, benign growths, or malignant tumors. Size, shape, growth over time, and patient risk factors help determine how concerning the nodule may be. Very small nodules are often monitored with repeat imaging rather than treated immediately.

Unlike a mass, a nodule is smaller. However, even small nodules may require careful follow-up depending on imaging features and clinical history.

How is it detected on a Chest X-ray?

Chest X-ray may show a pulmonary nodule as a small round or oval white spot within the lung field. Some nodules are difficult to see on standard X-rays, especially if they are tiny or hidden behind ribs, the heart, or diaphragm.

Radiologists evaluate:

- Size of the nodule
- Shape (round, oval, irregular)
- Borders (smooth, lobulated, spiculated)
- Density or calcification pattern
- Single nodule versus multiple nodules
- Stability compared with older scans
- Location (upper lobe nodules may receive closer attention)

Common findings include:

- Solitary pulmonary nodule (single spot)
- Multiple nodules scattered in one or both lungs
- Calcified nodule suggesting prior healed infection in some cases
- Spiculated margins that may raise concern for malignancy

CT scan is much more sensitive than chest X-ray and is usually used to evaluate nodules more accurately.

Symptoms

Most pulmonary nodules cause **no symptoms**, especially when small. They are commonly found incidentally during imaging done for cough, injury, surgery clearance, or other reasons.

When symptoms occur, they are often due to the underlying cause rather than the nodule itself.

Possible symptoms include:

- Persistent cough
- Mild shortness of breath
- Chest discomfort
- Wheezing if near an airway
- Recurrent infections in the same area
- Fatigue

If infection is the cause:

- Fever
- Night sweats
- Weight loss
- Productive cough

If cancer is present:

- Persistent cough
- Coughing blood
- Unexplained weight loss
- Reduced appetite
- Progressive fatigue
- Ongoing chest pain

Many patients remain symptom-free even when follow-up is required.

Causes & Risk Factors

Pulmonary nodules can result from many benign or malignant processes.

Benign Causes

- Healed granulomas from previous infections
- Tuberculosis exposure
- Fungal infections
- Hamartoma (benign tumor)
- Scar tissue from old inflammation
- Rheumatoid nodules or autoimmune disease
- Small areas of rounded atelectasis

Malignant Causes

- Early lung cancer
- Metastatic cancer spread from another organ
- Carcinoid tumors
- Lymphoma (less common)

Infectious / Inflammatory Causes

- Active tuberculosis
- Bacterial or fungal infection
- Sarcoidosis
- Vasculitis

Risk Factors for Concerning Nodules

- Smoking history
- Older age
- Prior cancer
- Family history of lung cancer
- Exposure to asbestos, radon, silica, or pollutants
- Growth seen on repeat imaging
- Larger size (>8 mm often more concerning than very small nodules)
- Irregular or spiculated borders

Prevention

Many nodules caused by old infections cannot be prevented, but cancer-related risk can often be reduced.

Reduce Lung Cancer Risk

- Do not smoke or stop smoking
- Avoid secondhand smoke
- Reduce occupational exposure to dust and toxins
- Test homes for radon where appropriate
- Exercise regularly and maintain healthy weight

Prevent Infectious Causes

- Seek treatment for tuberculosis or chronic lung infections
- Follow vaccination recommendations
- Use masks or respiratory protection in high-risk environments

Medical Follow-Up Prevention

- Do not ignore incidental nodules found on imaging
- Attend scheduled repeat CT scans if recommended
- Share prior imaging with doctors for comparison

High-Risk Screening

Eligible current or former smokers may benefit from low-dose CT lung cancer screening programs.

When to see a doctor

Any newly found pulmonary nodule should be reviewed by a healthcare professional to determine whether monitoring or further testing is needed.

See a doctor promptly if you have:

- Persistent cough lasting more than a few weeks
- Unexplained weight loss
- Ongoing fatigue
- Recurrent chest infections
- New shortness of breath
- Abnormal chest X-ray mentioning a nodule
- History of smoking with new respiratory symptoms

Seek urgent or emergency care if you develop:

- Coughing blood
- Severe shortness of breath
- Rapid worsening chest pain
- High fever with breathing symptoms
- Sudden weakness or collapse

People with smoking history, prior cancer, enlarging nodules, or suspicious CT findings should follow up quickly because early diagnosis of lung cancer offers the best treatment outcomes.

7. Pneumonia

What is it?

Pneumonia is an **infection of the lungs** that causes inflammation of the air sacs (alveoli). These air sacs may fill with fluid, pus, mucus, and inflammatory material, making it harder for oxygen to pass into the bloodstream. Pneumonia can affect one small area of the lung, an entire lobe, or both lungs.

It may be caused by bacteria, viruses, fungi, or aspiration of food or vomit into the lungs. Pneumonia ranges from mild illness that improves at home to life-threatening disease requiring hospitalization, oxygen support, or intensive care.

Young children, older adults, people with chronic diseases, smokers, and individuals with weakened immune systems are at higher risk for serious pneumonia.

How is it detected on a Chest X-ray?

Chest X-ray is one of the most common tests used to diagnose pneumonia. Radiologists look for areas where normal dark air-filled lung becomes whiter due to infection-related fluid and inflammation.

Common findings include:

- Focal or patchy white opacities in the lung
- **Lobar consolidation** involving one lobe
- Bronchopneumonia pattern with scattered patchy infiltrates
- Bilateral diffuse opacities in viral or severe pneumonia
- Air bronchograms (dark air-filled bronchi within white consolidated lung)
- Pleural effusion in some cases
- Lower lobe infiltrates in aspiration pneumonia

Sometimes early pneumonia may not be obvious on X-ray, especially if dehydration is present. CT scan may be used when suspicion remains high despite a normal X-ray.

Symptoms

Symptoms vary by age, cause, and severity. Some people become sick quickly, while others worsen gradually over several days.

Common symptoms include:

- Fever
- Chills or shivering
- Cough, often producing mucus
- Yellow, green, rust-colored, or blood-tinged sputum
- Shortness of breath
- Rapid breathing
- Chest pain that worsens with coughing or deep breathing
- Fatigue and weakness
- Sweating
- Loss of appetite

Symptoms in children may include:

- Fast breathing
- Poor feeding
- Irritability
- Vomiting
- Lethargy
- Grunting or chest retractions

Symptoms in older adults may include:

- Confusion
- Sudden weakness
- Poor appetite
- Sleepiness
- Falls
- Mild or absent fever despite serious infection

Severe pneumonia may cause:

- Low oxygen levels
- Bluish lips

- Dehydration
- Severe breathlessness
- Inability to speak full sentences
- Collapse or altered mental status

Causes & Risk Factors

Pneumonia occurs when germs or other material enter the lungs and overwhelm normal defenses.

Infectious Causes

Bacterial Pneumonia

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*
- *Staphylococcus aureus*
- Atypical bacteria such as *Mycoplasma pneumoniae*

Viral Pneumonia

- Influenza viruses
- RSV
- Coronaviruses
- Other respiratory viruses

Fungal Pneumonia

More common in immunocompromised individuals.

Aspiration Pneumonia

Occurs when food, liquid, saliva, or vomit enters the lungs.

Risk Factors

- Age over 65 or very young children
- Smoking
- COPD, asthma, or chronic lung disease
- Diabetes
- Heart disease
- Kidney or liver disease
- Cancer
- Weak immune system

- Recent viral illness
- Alcohol misuse
- Poor nutrition
- Difficulty swallowing
- Bedridden state or recent hospitalization

Hospital-acquired pneumonia can occur in hospitalized or ventilated patients.

Prevention

Many cases of pneumonia can be prevented or the severity reduced.

Vaccination

- Influenza vaccine
- Pneumococcal vaccines when eligible
- Other recommended respiratory vaccines based on age/risk

Lifestyle Prevention

- Do not smoke or stop smoking
- Maintain healthy nutrition
- Exercise regularly
- Get adequate sleep
- Limit alcohol misuse

Infection Control

- Frequent handwashing
- Avoid close contact with sick individuals when possible
- Cover coughs and sneezes
- Stay home when ill to reduce spread

Chronic Disease Management

- Control asthma, COPD, diabetes, and heart disease
- Use inhalers properly when prescribed

Aspiration Prevention

- Sit upright while eating
- Manage swallowing problems
- Oral hygiene, especially in elderly or bedridden patients

When to see a doctor

Medical attention is important if pneumonia is suspected, especially in vulnerable individuals.

See a doctor promptly if you have:

- Fever with cough lasting more than a few days
- Shortness of breath
- Chest pain while breathing
- Worsening cough with mucus
- Fatigue that is severe or persistent
- Symptoms after influenza or viral illness
- Pneumonia symptoms in elderly adults or young children

Seek urgent or emergency care if you develop:

- Severe breathing difficulty
- Rapid breathing
- Blue lips or fingertips
- Low oxygen saturation
- Confusion or unusual sleepiness
- Inability to drink fluids
- Persistent vomiting
- Coughing blood
- High fever with weakness
- Chest pain or collapse

Older adults, infants, pregnant individuals, and people with weakened immunity should seek care early because pneumonia can worsen quickly and may become life-threatening without treatment.

8. Pneumothorax

What is it?

Pneumothorax is a **collapsed lung caused by air leaking into the pleural space**, the space between the lung and the chest wall. When air enters this space, it reduces the normal negative pressure that keeps the lung expanded, causing part or all of the lung to collapse.

Pneumothorax can occur suddenly and may range from a small partial collapse to a life-threatening emergency. Some cases resolve on their own, while others require urgent removal of air with a needle or chest tube.

There are several types:

- **Spontaneous pneumothorax:** occurs without trauma, often from rupture of small air blisters (blebs)
- **Secondary pneumothorax:** occurs in people with underlying lung disease such as COPD
- **Traumatic pneumothorax:** caused by injury to the chest
- **Iatrogenic pneumothorax:** caused by medical procedures
- **Tension pneumothorax:** dangerous form where pressure builds rapidly and compresses the heart and lungs

How is it detected on a Chest X-ray?

Chest X-ray is one of the primary tests used to diagnose pneumothorax. Radiologists look for separation between the lung and chest wall caused by trapped air.

Common findings include:

- Visible thin **pleural line** representing the edge of the collapsed lung
- No lung markings beyond that pleural line
- Increased blackness (air) in the pleural space
- Partial or complete collapse of lung tissue toward the center
- Flattened diaphragm in larger cases
- Mediastinal shift away from the affected side in tension pneumothorax
- Depressed diaphragm and widened rib spaces in severe cases

Small pneumothoraces may be subtle on X-ray. CT scan can detect very small or uncertain cases.

Symptoms

Symptoms usually begin suddenly, though severity depends on size of the collapse and underlying lung health.

Common symptoms include:

- Sudden sharp chest pain, often one-sided
- Shortness of breath
- Chest tightness
- Rapid breathing
- Dry cough
- Anxiety or feeling unable to get enough air

Other possible symptoms:

- Fast heart rate
- Fatigue
- Shoulder pain
- Reduced exercise tolerance

In people with COPD or other lung disease, even a small pneumothorax may cause severe symptoms.

Signs of Severe or Tension Pneumothorax

- Extreme shortness of breath
- Severe chest pain
- Bluish lips or fingertips
- Rapid heart rate
- Low blood pressure
- Confusion
- Collapse
- Distended neck veins (sometimes)

Tension pneumothorax is a medical emergency.

Causes & Risk Factors

Pneumothorax occurs when air leaks from the lung or enters through the chest wall.

Spontaneous Causes

- Rupture of small blebs near lung surface
- Often occurs in tall, thin young adults
- Smoking significantly increases risk

Secondary Causes (Underlying Lung Disease)

- COPD / emphysema
- Asthma (rarely severe attacks)
- Tuberculosis
- Pneumonia
- Cystic fibrosis
- Lung cancer
- Interstitial lung disease

Traumatic Causes

- Rib fractures
- Stab or gunshot wounds
- Car accidents
- Sports injuries
- Blast injury

Medical Procedure Causes

- Central line placement
- Lung biopsy
- Mechanical ventilation
- Thoracentesis

Risk Factors

- Smoking
- Previous pneumothorax
- Family history
- Male sex (for spontaneous type)
- Tall thin body habitus
- Chronic lung disease
- High-pressure ventilation
- Scuba diving or high-altitude pressure changes (in some cases)

Prevention

Not all pneumothoraces can be prevented, but recurrence risk and secondary causes can often be reduced.

Reduce Recurrence Risk

- Stop smoking completely
- Follow up after any previous pneumothorax
- Avoid air travel or diving until medically cleared
- Follow surgeon advice after pleurodesis or repair procedures

Manage Lung Disease

- Proper treatment of COPD or asthma

- Vaccination and infection prevention
- Seek treatment early for pneumonia or severe cough

Safety Measures

- Use seatbelts and injury prevention strategies
- Protective gear in contact sports or hazardous work

Medical Setting Prevention

- Careful technique during chest procedures
- Monitoring high-risk ventilated patients

When to see a doctor

Any sudden chest pain or unexplained shortness of breath should be medically evaluated.

See a doctor promptly if you have:

- Sudden one-sided chest pain
- Mild shortness of breath
- Persistent chest tightness
- New symptoms after lung procedure
- Previous pneumothorax with similar symptoms

Seek urgent or emergency care immediately if you develop:

- Severe breathing difficulty
- Rapid worsening shortness of breath
- Blue lips or low oxygen levels
- Severe chest pain
- Fast heart rate with dizziness
- Fainting or collapse
- Symptoms after chest injury
- Increasing distress after known pneumothorax

People with COPD, emphysema, prior pneumothorax, or recent chest trauma should seek immediate care because the condition can worsen rapidly and may become life-threatening.

9. Consolidation

What is it?

Consolidation is a **radiology term describing lung tissue that has become filled with material such as fluid, pus, blood, cells, or protein instead of normal air**. When the tiny air sacs (alveoli) are filled, the affected part of the lung becomes denser and less able to exchange oxygen efficiently.

Consolidation is **not a disease itself**, but an imaging finding that usually indicates an underlying condition. The most common cause is pneumonia, but consolidation can also occur from pulmonary edema, bleeding into the lungs, aspiration, inflammatory disease, or certain cancers.

It may affect a small segment, an entire lobe, or multiple areas in one or both lungs. Depending on the cause and size of involvement, consolidation may cause mild symptoms or severe respiratory distress.

How is it detected on a Chest X-ray?

Chest X-ray commonly reveals consolidation as a dense white area replacing the normal dark air-filled appearance of the lung.

Typical findings include:

- Homogeneous or patchy white opacity in a lung segment or lobe
- Loss of normal visibility of nearby structures (silhouette sign)
- **Air bronchograms**: dark branching air-filled bronchi visible within white consolidated lung
- Lobar consolidation with sharp fissural boundaries
- Multifocal patchy consolidation in bronchopneumonia
- Bilateral consolidation in severe infection or edema
- Lower lobe involvement in aspiration-related disease

Radiologists also assess location, extent, and associated findings such as pleural effusion, enlarged heart, or cavitation to help identify the cause.

CT scan may be used when chest X-ray findings are unclear or persistent.

Symptoms

Symptoms depend on the underlying cause, extent of lung involvement, and patient health status.

Common symptoms include:

- Cough, dry or productive
- Fever or chills (especially if infection is present)
- Shortness of breath

- Rapid breathing
- Chest pain, especially with deep breathing or coughing
- Fatigue
- Weakness
- Reduced exercise tolerance

If pneumonia is the cause:

- Thick mucus or sputum
- Sweating
- Loss of appetite
- Body aches

If fluid overload or edema is the cause:

- Breathlessness when lying flat
- Leg swelling
- Sudden nighttime shortness of breath

If bleeding is the cause:

- Coughing blood
- Severe breathlessness
- Dizziness or weakness

In older adults:

- Confusion
- Poor appetite
- Sudden weakness
- Falls
- Sleepiness

Severe cases may lead to:

- Low oxygen levels
- Blue lips
- Inability to speak full sentences
- Respiratory distress

Causes & Risk Factors

Consolidation can result from several lung disorders.

Common Causes

Infection

- Bacterial pneumonia
- Viral pneumonia
- Fungal infections
- Tuberculosis (sometimes)

Fluid / Edema

- Heart failure causing pulmonary edema

Aspiration

- Food, vomit, or liquid entering the lungs

Hemorrhage

- Bleeding into lung tissue

Inflammatory Conditions

- Autoimmune lung disease
- Organizing pneumonia
- Eosinophilic lung disease

Tumor-related Causes

- Certain lung cancers that mimic pneumonia
- Tumor blocking airway causing post-obstructive infection

Risk Factors

- Smoking
- Older age
- Chronic lung disease (COPD, asthma)
- Diabetes
- Heart disease
- Weak immune system
- Difficulty swallowing
- Bedridden state
- Recent surgery
- Alcohol misuse

- Prior lung infection

Prevention

Because consolidation is a finding rather than a single disease, prevention focuses on common causes.

Prevent Infections

- Vaccinations such as influenza and pneumococcal vaccines when appropriate
- Hand hygiene
- Seek treatment early for cough, fever, or respiratory infection
- Avoid smoking

Prevent Aspiration

- Sit upright while eating
- Manage swallowing problems
- Avoid heavy sedation when possible
- Good oral hygiene

Prevent Heart-Related Causes

- Control blood pressure
- Manage heart failure carefully
- Limit excess salt if advised
- Take medications as prescribed

Protect Lung Health

- Exercise regularly
- Manage COPD or asthma
- Avoid harmful dust, fumes, and smoke exposure

Follow-Up Care

Persistent consolidation on repeat imaging should always be re-evaluated to rule out hidden obstruction or cancer.

When to see a doctor

Medical evaluation is recommended for symptoms suggesting lung infection or breathing impairment.

See a doctor promptly if you have:

- Persistent cough lasting more than a few weeks

- Fever with cough
- New shortness of breath
- Chest pain while breathing
- Fatigue that is worsening
- Recurrent infections
- Abnormal chest X-ray showing consolidation

Seek urgent or emergency care if you develop:

- Severe breathing difficulty
- Rapid breathing
- Blue lips or low oxygen levels
- Confusion
- Coughing blood
- High fever with weakness
- Inability to eat or drink
- Collapse or fainting

Older adults, immunocompromised individuals, and people with heart or lung disease should seek care early because consolidation may represent serious pneumonia or another urgent lung condition.

10. Edema (Pulmonary Edema)

What is it?

Pulmonary edema is the **buildup of excess fluid inside the lungs**, especially within the air sacs (alveoli) and surrounding lung tissue. Instead of remaining dry and filled with air, the alveoli become flooded with fluid, making it difficult for oxygen to move into the bloodstream. This can cause rapid breathing difficulty and may become life-threatening if not treated quickly.

Pulmonary edema is most commonly caused by **heart problems**, especially heart failure, where the heart cannot pump blood efficiently and pressure backs up into the lungs. It can also occur from kidney failure, severe infections, inhalation injury, trauma, high altitude exposure, or direct lung injury.

Pulmonary edema may develop suddenly (acute pulmonary edema), which is a medical emergency, or gradually over time in chronic heart disease.

How is it detected on a Chest X-ray?

Chest X-ray is one of the key tests used to identify pulmonary edema. Radiologists look for signs of fluid accumulation in the lungs and often evidence of heart enlargement.

Common findings include:

- Bilateral diffuse white or hazy lung opacities
- “Bat-wing” or central perihilar fluffy opacities in cardiogenic edema
- Interstitial edema with fine linear markings (Kerley B lines)
- Enlarged heart silhouette if heart failure is the cause
- Fluid in lung fissures
- Pleural effusions, often on both sides
- Vascular congestion or prominent upper lung blood vessels
- Lower lung opacities in some cases

The pattern helps distinguish heart-related edema from other lung diseases. CT scan or ultrasound may also support diagnosis.

Symptoms

Symptoms depend on how quickly fluid accumulates and the underlying cause.

Common Symptoms

- Shortness of breath
- Difficulty breathing during activity
- Feeling unable to take a deep breath
- Rapid breathing
- Cough
- Fatigue
- Reduced exercise tolerance

Symptoms of Heart-Related or Chronic Edema

- Shortness of breath when lying flat (orthopnea)
- Waking suddenly at night gasping for air
- Swelling of ankles or legs
- Weight gain from fluid retention
- Persistent cough, especially at night

Acute Severe Symptoms

- Sudden severe breathlessness

- Pink frothy sputum
- Wheezing or gurgling breathing sounds
- Sweating
- Anxiety or panic sensation
- Fast heartbeat
- Blue lips or fingertips
- Extreme fatigue

In Older Adults

- Confusion
- Weakness
- Sudden decline in walking ability
- Sleepiness

Acute pulmonary edema requires emergency treatment.

Causes & Risk Factors

Pulmonary edema happens when fluid leaks from blood vessels into lung tissue.

Cardiogenic Causes (Most Common)

Due to increased pressure from heart disease:

- Congestive heart failure
- Heart attack
- Cardiomyopathy
- Severe high blood pressure
- Valve disease
- Dangerous arrhythmias

Non-Cardiogenic Causes

Fluid leak due to lung injury or other illness:

- Severe infection / sepsis
- Acute respiratory distress syndrome (ARDS)
- Kidney failure with fluid overload
- Near drowning
- Smoke or toxic inhalation

- Head injury or neurological events
- High altitude pulmonary edema
- Drug overdose or reactions

Risk Factors

- Known heart disease
- High blood pressure
- Diabetes
- Obesity
- Smoking
- Kidney disease
- Older age
- Prior heart attack
- Valve disease
- Poor medication adherence in heart failure

Prevention

Many cases can be prevented by managing heart disease and overall health.

Heart Health Prevention

- Control blood pressure
- Take heart failure medicines regularly
- Limit excess salt if advised
- Monitor weight for sudden fluid gain
- Exercise as medically appropriate
- Maintain healthy weight
- Manage cholesterol and diabetes

Lifestyle Prevention

- Stop smoking
- Limit alcohol misuse
- Reduce illicit drug use
- Get regular sleep and manage stress

Prevent Non-Cardiac Causes

- Treat infections early
- Follow kidney disease treatment plans
- Use oxygen and acclimatization precautions at high altitude if at risk
- Use protective equipment around toxic fumes

Follow-Up Care

People with known heart failure should attend regular appointments and seek early care for worsening symptoms.

When to see a doctor

Because pulmonary edema can worsen quickly, breathing symptoms should never be ignored.

See a doctor promptly if you have:

- Increasing shortness of breath with activity
- New difficulty lying flat
- Leg swelling
- Sudden weight gain over a few days
- Persistent cough at night
- Fatigue that is worsening
- Known heart disease with new breathing symptoms

Seek urgent or emergency care immediately if you develop:

- Sudden severe shortness of breath
- Gasping for air
- Pink frothy sputum
- Blue lips or fingertips
- Chest pain
- Rapid heartbeat with weakness
- Confusion
- Inability to speak full sentences
- Collapse or fainting

People with heart failure, kidney disease, prior heart attack, or severe hypertension should seek urgent evaluation early because pulmonary edema can become life-threatening very rapidly.

11. Emphysema

What is it?

Emphysema is a **chronic lung disease in which the tiny air sacs (alveoli) in the lungs become damaged and enlarged**, reducing the surface area needed for oxygen exchange. The walls between alveoli break down over time, creating larger but less efficient air spaces. This makes it harder to breathe out fully and traps air inside the lungs.

Emphysema is one of the main conditions included under **chronic obstructive pulmonary disease (COPD)**. It usually develops slowly over many years and is most strongly associated with smoking. Once lung tissue is destroyed, the damage is generally permanent, but treatment can slow progression and improve quality of life.

People with emphysema may gradually lose exercise capacity, experience chronic breathlessness, and become more vulnerable to lung infections and pneumothorax.

How is it detected on a Chest X-ray?

Chest X-ray may suggest emphysema, especially in moderate to advanced disease. Radiologists look for signs of hyperinflation and lung tissue destruction.

Common findings include:

- Overexpanded lungs with increased lung volume
- Flattened diaphragms
- Increased blackness of the lungs due to trapped air
- Widened spaces between ribs
- Long, narrow heart silhouette from hyperinflation
- Reduced visible vascular markings in peripheral lungs
- Large air spaces called **bullae** in some patients
- Increased retrosternal airspace on lateral views

Early emphysema may not be obvious on X-ray. CT scan is more sensitive and can better show the extent and type of emphysema.

Symptoms

Symptoms usually develop gradually and worsen over time.

Common Symptoms

- Shortness of breath, especially during exertion
- Difficulty climbing stairs or walking uphill

- Chronic cough (sometimes with chronic bronchitis overlap)
- Wheezing
- Chest tightness
- Fatigue
- Reduced exercise tolerance

Progressive Symptoms

- Breathlessness during routine activities
- Prolonged exhalation
- Frequent respiratory infections
- Need to pause while walking
- Weight loss due to increased work of breathing

Advanced Symptoms

- Shortness of breath at rest
- Use of neck or chest muscles to breathe
- Barrel-shaped chest from chronic hyperinflation
- Low oxygen levels
- Blue lips or fingertips
- Swelling of legs if heart strain develops

During Flare-Ups (Exacerbations)

- Sudden worsening cough
- Increased mucus
- More wheezing
- Marked breathlessness

Causes & Risk Factors

Emphysema develops when lung tissue is repeatedly damaged.

Main Causes

Smoking (Most Common Cause)

- Cigarette smoking
- Cigar or pipe smoking
- Long-term tobacco exposure

Other Causes

- Secondhand smoke exposure
- Occupational dusts, fumes, and chemicals
- Air pollution
- Biomass smoke exposure (indoor cooking smoke)

Genetic Cause

- **Alpha-1 antitrypsin deficiency**, a hereditary condition that can cause emphysema at a younger age

Risk Factors

- Long smoking history
- Age over 40 (commonly diagnosed later)
- Chronic bronchitis or COPD history
- Recurrent respiratory infections
- Occupational inhalation exposure
- Family history of COPD or alpha-1 antitrypsin deficiency
- Poor air quality exposure

Prevention

Emphysema is often preventable, especially when tobacco exposure is avoided.

Most Important Prevention

- Do not smoke
- Stop smoking as early as possible if currently smoking
- Avoid secondhand smoke

Protect the Lungs

- Use masks or respirators around dust and fumes
- Improve ventilation in cooking environments
- Reduce exposure to pollution when possible

Maintain Lung Health

- Stay physically active
- Keep vaccinations current (influenza, pneumococcal when appropriate)
- Treat respiratory infections early

- Manage asthma or chronic bronchitis

Screening / Family Risk

- Discuss testing for alpha-1 antitrypsin deficiency if emphysema occurs at a young age or without smoking history

Medical Follow-Up

Early diagnosis allows inhalers, pulmonary rehabilitation, and smoking cessation support that can slow progression.

When to see a doctor

Medical evaluation is recommended for chronic breathing symptoms, especially in smokers or former smokers.

See a doctor promptly if you have:

- Increasing shortness of breath
- Chronic cough
- Wheezing
- Frequent chest infections
- Reduced ability to exercise
- Need to stop frequently while walking
- Unexplained weight loss

Seek urgent or emergency care if you develop:

- Severe breathing difficulty
- Rapid worsening during a flare-up
- Blue lips or low oxygen readings
- Confusion or extreme fatigue
- Chest pain
- Sudden one-sided chest pain (possible pneumothorax)
- Inability to speak full sentences
- Collapse

Smokers, former smokers, and people with known COPD should seek early care because treatment can reduce symptoms, prevent flare-ups, and slow further lung damage.

12. Fibrosis (Pulmonary Fibrosis)

What is it?

Pulmonary fibrosis is a **chronic lung condition in which lung tissue becomes scarred, thickened, and stiff** over time. The scarring affects the supporting tissue around the air sacs, making it harder for the lungs to expand normally and harder for oxygen to pass into the bloodstream.

Unlike infections that may heal completely, fibrosis usually represents long-term structural damage. In many cases the scarring gradually worsens, leading to increasing breathlessness and reduced lung function. Some forms progress slowly over years, while others worsen more rapidly.

Pulmonary fibrosis may occur without a known cause (idiopathic pulmonary fibrosis), or it may develop due to autoimmune disease, environmental exposures, prior infections, radiation, certain medications, or chronic inflammatory lung disease.

Because scarring is often permanent, early diagnosis and treatment are important to slow progression and preserve quality of life.

How is it detected on a Chest X-ray?

Chest X-ray may suggest fibrosis, especially in moderate or advanced disease, although CT scan is more accurate.

Common X-ray findings include:

- Reticular (net-like) or streaky white opacities
- Increased interstitial markings
- Volume loss with smaller appearing lungs
- Predominantly lower lung or peripheral abnormalities in some types
- Irregular scarring patterns
- Elevated diaphragms due to reduced lung volume
- Distortion of normal lung architecture

In advanced disease:

- Honeycomb-like changes may be suspected (better seen on CT)
- Traction bronchiectasis may be present on CT
- Bilateral chronic fibrotic changes

High-resolution CT is usually the preferred imaging test to define the pattern and extent of fibrosis.

Symptoms

Symptoms usually develop gradually, though some patients experience more rapid decline.

Common Symptoms

- Progressive shortness of breath, especially with exertion
- Dry persistent cough
- Fatigue
- Reduced stamina
- Difficulty climbing stairs or walking long distances

Ongoing / Progressive Symptoms

- Breathlessness during routine daily activities
- Need to stop frequently while walking
- Poor exercise tolerance
- Chest discomfort or tightness
- Weight loss in some cases

Advanced Symptoms

- Shortness of breath at rest
- Low oxygen levels
- Need for supplemental oxygen
- Bluish lips or fingertips
- Clubbing (rounded fingertips) in some patients
- Severe fatigue and weakness

During Acute Worsening

- Sudden increase in breathlessness
- New cough
- Low oxygen levels
- Fever if infection is present

Older adults may attribute symptoms to aging until disease becomes advanced.

Causes & Risk Factors

Fibrosis occurs when repeated lung injury or abnormal healing leads to scar formation.

Idiopathic Causes

- **Idiopathic pulmonary fibrosis (IPF):** no clear known cause

Autoimmune / Connective Tissue Causes

- Rheumatoid arthritis
- Scleroderma
- Lupus
- Myositis-related disease

Environmental / Occupational Causes

- Asbestos exposure
- Silica dust
- Coal dust
- Bird proteins or mold (hypersensitivity pneumonitis)
- Farming or industrial exposures

Medical Causes

- Prior severe pneumonia
- Tuberculosis scarring
- Radiation therapy to the chest
- Certain medications (some chemotherapy drugs, amiodarone, others)
- Sarcoidosis

Risk Factors

- Older age
- Smoking
- Family history of fibrosis
- Occupational inhalation exposures
- Autoimmune disease
- Male sex in some forms
- Gastroesophageal reflux may coexist in some patients

Prevention

Not all fibrosis can be prevented, especially idiopathic forms, but risk from known causes can often be reduced.

Lung Protection

- Do not smoke or stop smoking
- Use masks/respiratory protection around dust, mold, or chemicals

- Follow workplace safety standards
- Improve ventilation in exposure-prone environments

Medical Prevention

- Treat autoimmune diseases appropriately
- Monitor medications that may affect lungs
- Seek treatment early for severe lung infections
- Follow up after radiation or chemotherapy when advised

General Health

- Stay active within tolerance
- Maintain healthy nutrition
- Vaccinations such as influenza and pneumococcal vaccines when appropriate
- Avoid respiratory infections when possible

Early Detection

People with chronic dry cough or unexplained exertional breathlessness should seek evaluation early, as earlier treatment may slow progression.

When to see a doctor

Medical review is important for unexplained chronic breathlessness or cough.

See a doctor promptly if you have:

- Increasing shortness of breath over weeks or months
- Dry cough that persists
- Reduced exercise tolerance
- Fatigue without clear cause
- Crackling sounds in lungs noted by a clinician
- Unexplained weight loss
- Abnormal chest X-ray suggesting scarring

Seek urgent or emergency care if you develop:

- Sudden worsening breathlessness
- Low oxygen levels
- Blue lips or confusion
- Fever with breathing decline

- Chest pain
- Inability to walk or speak comfortably due to breathlessness
- Collapse or severe weakness

People with known pulmonary fibrosis should seek early care for any sudden change, as infections or acute exacerbations can become serious quickly.

13. Pleural Thickening

What is it?

Pleural thickening is a **thickening or scarring of the pleura**, the thin membrane that lines the lungs and the inside of the chest wall. Normally, the pleura is very thin and smooth, allowing the lungs to move easily during breathing. When it becomes thickened, stiff, or scarred, lung movement may be restricted and normal breathing mechanics can be affected.

Pleural thickening is usually a **radiologic finding rather than a disease itself**. It often develops after prior inflammation, infection, injury, surgery, bleeding, or long-term exposure to irritants such as asbestos. In some people it causes no symptoms and is discovered incidentally on imaging. In others, especially when extensive, it can contribute to breathlessness, chest discomfort, or reduced lung function.

Pleural thickening may affect one small area (focal thickening) or involve larger sections of the pleura (diffuse pleural thickening).

How is it detected on a Chest X-ray?

Chest X-ray may show pleural thickening as abnormal dense lines or areas along the edge of the lung or chest wall.

Common findings include:

- White linear or plaque-like thickening along the chest wall
- Blunting or irregularity of the costophrenic angle
- Apical pleural capping (thickening at the lung tops)
- Reduced sharpness of the pleural lining
- Calcified pleural plaques in asbestos-related disease
- Volume loss if extensive scarring restricts lung expansion
- Associated old pleural effusion or prior rib/chest abnormalities

Because subtle pleural thickening can be difficult to fully assess on X-ray, **CT scan is more accurate** and commonly used for confirmation.

Symptoms

Many people with mild pleural thickening have no symptoms. Symptoms are more likely when thickening is extensive or associated with another lung condition.

Common Symptoms

- Mild shortness of breath
- Reduced ability to exercise
- Chest tightness
- Dull chest discomfort
- Fatigue with exertion

If Lung Restriction Develops

- Difficulty taking a deep breath
- Breathlessness during walking or climbing stairs
- Reduced stamina
- Persistent dry cough

If Related to Ongoing Inflammation or Other Disease

- Sharp pain with breathing
- Fever (if infection present)
- Weight loss (if malignancy or chronic disease is present)

Advanced / Extensive Cases

- Significant breathlessness
- Low oxygen levels during exertion
- Marked reduction in exercise capacity

Symptoms often depend more on the underlying cause than the thickening itself.

Causes & Risk Factors

Pleural thickening usually develops after injury, inflammation, or chronic irritation of the pleura.

Common Causes

Previous Infection

- Tuberculosis
- Severe pneumonia involving pleura

- Empyema (infected pleural fluid)

Prior Fluid or Bleeding

- Pleural effusion
- Hemothorax (blood in pleural space)

Occupational / Environmental Exposure

- **Asbestos exposure** (important cause of pleural plaques and thickening)

Inflammatory Causes

- Autoimmune diseases such as rheumatoid arthritis or lupus
- Chronic pleuritis

Trauma / Procedures

- Chest injury
- Thoracic surgery
- Repeated pleural drainage procedures

Tumor-Related Causes

- Mesothelioma
- Pleural metastases
- Other cancers involving pleura

Risk Factors

- Occupational exposure to asbestos, insulation, shipyards, construction dust
- Prior tuberculosis or chest infection
- Smoking (especially with asbestos increases lung cancer risk)
- Prior chest trauma or surgery
- Older age
- Chronic inflammatory disease

Prevention

Some causes can be prevented or risks reduced.

Occupational Prevention

- Avoid asbestos exposure
- Use approved protective equipment in hazardous workplaces
- Follow workplace safety regulations

- Report unsafe environmental exposures

Infection Prevention

- Early treatment of pneumonia and tuberculosis
- Vaccination when appropriate
- Seek treatment for pleural infections promptly

General Lung Health

- Do not smoke or stop smoking
- Maintain fitness and breathing capacity through activity
- Avoid repeated exposure to dust or fumes

Medical Follow-Up

- Follow up previous pleural effusions or chest infections
- Attend imaging reviews if pleural abnormalities are found
- Monitor long-term asbestos exposure history with a healthcare provider

When to see a doctor

Medical review is recommended if pleural thickening is newly found or causing symptoms.

See a doctor promptly if you have:

- New or worsening shortness of breath
- Persistent chest discomfort
- Ongoing dry cough
- Reduced exercise tolerance
- History of asbestos exposure with respiratory symptoms
- Abnormal chest X-ray mentioning pleural thickening

Seek urgent or emergency care if you develop:

- Severe breathing difficulty
- Sharp chest pain with shortness of breath
- Fever with chest symptoms
- Coughing blood
- Rapid decline in stamina or oxygen levels
- Confusion or collapse

People with asbestos exposure, unexplained weight loss, or progressive symptoms should seek timely evaluation because further testing may be needed to rule out serious pleural disease.

14. Hernia (Hiatal Hernia / Diaphragmatic Hernia on Chest X-ray)

What is it?

In chest X-ray datasets, “hernia” most commonly refers to a **hiatal hernia** or occasionally a **diaphragmatic hernia**, where part of the stomach or abdominal organs moves upward through an opening or weakness in the diaphragm into the chest cavity.

The diaphragm is the muscle that separates the chest from the abdomen and helps with breathing. If abdominal contents move through the diaphragm, they may appear as an abnormal structure in the lower chest on imaging. A small hiatal hernia may cause no symptoms, while larger hernias can lead to reflux, swallowing difficulty, chest discomfort, or breathing problems.

Common Types

Hiatal Hernia

Part of the stomach slides upward through the opening where the esophagus passes through the diaphragm. This is the most common type seen in adults.

Paraesophageal Hernia

Part of the stomach pushes beside the esophagus into the chest and may carry greater risk of complications.

Diaphragmatic Hernia

Abdominal organs move into the chest through a defect in the diaphragm. This may be congenital or due to trauma.

How is it detected on a Chest X-ray?

Chest X-ray may suggest a hernia when structures that normally belong in the abdomen appear in the chest.

Common findings include:

- Air-fluid level behind the heart (retrocardiac region), often from stomach in hiatal hernia
- Rounded mass-like opacity in lower chest
- Gas bubble above the diaphragm
- Elevated or irregular diaphragm contour
- Bowel loops visible in the chest in diaphragmatic hernia
- Shift of nearby lung or mediastinal structures if large

Hiatal hernias are often better confirmed with:

- Barium swallow study
- CT scan
- Endoscopy

Radiologists correlate imaging findings with symptoms and other tests.

Symptoms

Many small hiatal hernias cause no symptoms and are found incidentally.

Common Symptoms (Especially Hiatal Hernia)

- Heartburn or acid reflux
- Sour taste in the mouth
- Regurgitation after meals
- Chest discomfort or burning
- Difficulty swallowing
- Feeling full quickly
- Belching or bloating

Respiratory or Chest Symptoms

- Chronic cough
- Hoarseness
- Wheezing from reflux irritation
- Shortness of breath, especially if large hernia compresses lungs
- Recurrent aspiration symptoms

Larger or Complicated Hernias

- Severe chest or upper abdominal pain
- Vomiting
- Inability to swallow
- Food getting stuck
- Anemia from chronic irritation or bleeding
- Sudden worsening reflux

In Infants / Congenital Cases

- Breathing distress
- Feeding problems

- Poor growth

Causes & Risk Factors

Hernias occur when tissue pushes through a natural opening or weakened area.

Hiatal Hernia Causes

- Weakening of tissues around the diaphragm opening with age
- Increased abdominal pressure
- Repeated straining

Diaphragmatic Hernia Causes

- Congenital defect present from birth
- Trauma such as motor vehicle accidents or severe injury
- Prior surgery

Risk Factors

- Older age
- Obesity
- Pregnancy
- Chronic coughing
- Heavy lifting
- Constipation with straining
- Smoking (can worsen cough and reflux)
- Connective tissue weakness
- Previous abdominal or chest surgery

Risk Factors for Symptoms

- Large meals
- Lying down soon after eating
- Alcohol use
- Spicy/fatty foods (reflux symptoms)
- Untreated obesity

Prevention

Some hernias cannot be fully prevented, but symptoms and risk may be reduced.

Reduce Pressure on the Abdomen

- Maintain healthy weight
- Avoid chronic straining during bowel movements
- Treat constipation
- Use proper lifting technique
- Avoid lifting excessive loads suddenly

Reduce Reflux Symptoms

- Eat smaller meals
- Avoid lying down for 2–3 hours after eating
- Elevate head of bed if nighttime reflux occurs
- Limit trigger foods if sensitive
- Reduce alcohol intake
- Stop smoking

General Health

- Treat chronic cough
- Stay active
- Follow up if known hernia enlarges or symptoms worsen

When to see a doctor

Medical evaluation is recommended if symptoms are frequent, worsening, or affecting eating and breathing.

See a doctor promptly if you have:

- Frequent heartburn or reflux
- Trouble swallowing
- Chest discomfort after meals
- Chronic cough or hoarseness
- Recurrent vomiting
- Feeling food gets stuck
- Abnormal chest X-ray suggesting hernia

Seek urgent or emergency care if you develop:

- Severe chest or upper abdominal pain

- Persistent vomiting
- Inability to swallow liquids or food
- Black stools or vomiting blood
- Sudden severe shortness of breath
- Rapid bloating with pain
- Collapse or severe weakness

People with large paraesophageal hernias may require earlier specialist review because strangulation or obstruction can occasionally become serious.